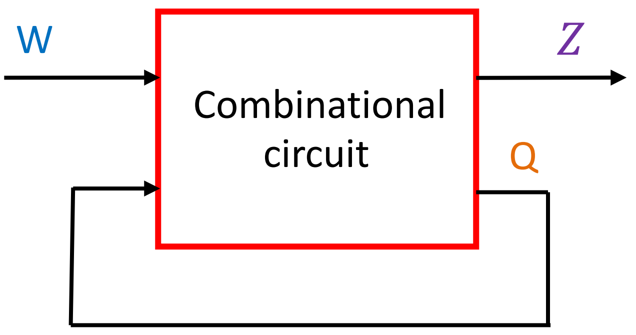
ECE 124 – Digital Circuits and Systems
Dept. of ECE, Univ. of Waterloo

Lecture Slides: **Set 7 - Part 1**

Contents:
- Introduction to asynchronous sequential circuits

Introduction

- An asynchronous sequential circuit (ASC)
  - does not have any clock input (i.e., no FFs) and
  - consists of combinational logic with feedback paths
- The feedback creates a memory effect (e.g., SR latch)
- General form of an ASC is shown below, where
  - W: circuit input
  - Q: Feedback or current state
  - Z: circuit output

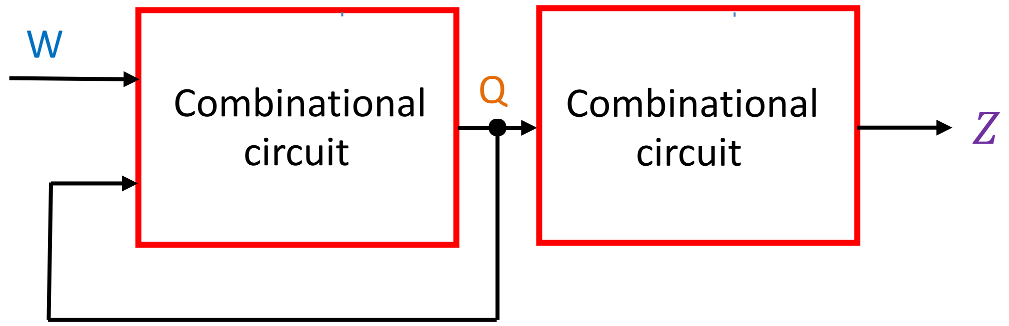


Note that W, Z and Q can each be of multiple bits

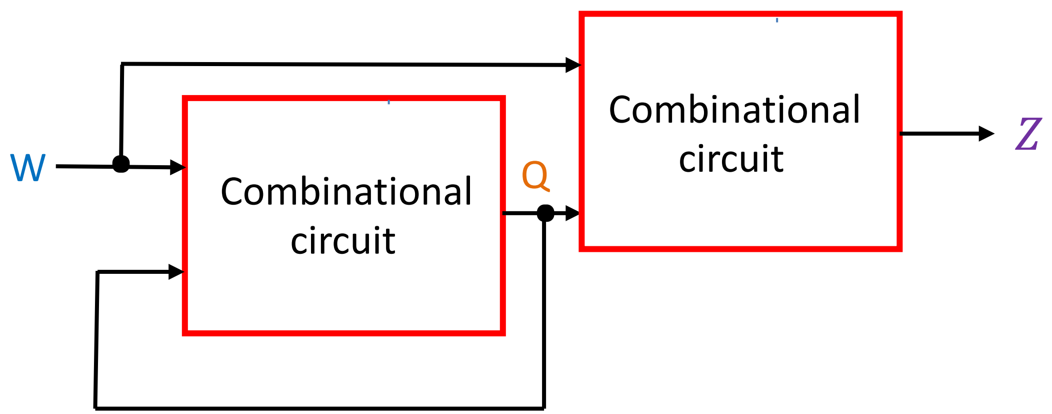
Introduction (contd.)

Like its synchronous counterpart, an asynchronous sequential circuit can be either Moore or Mealy type

Moore

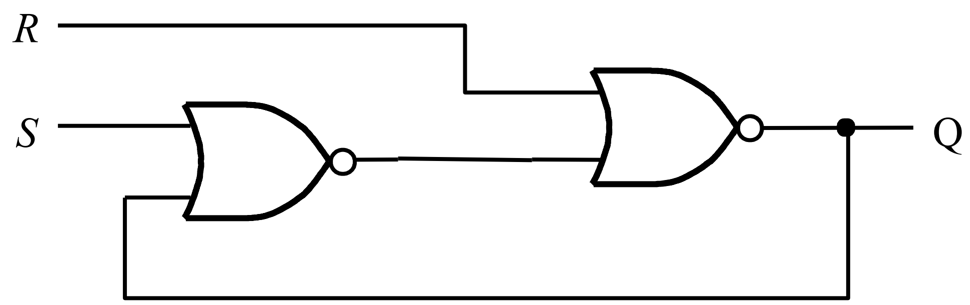


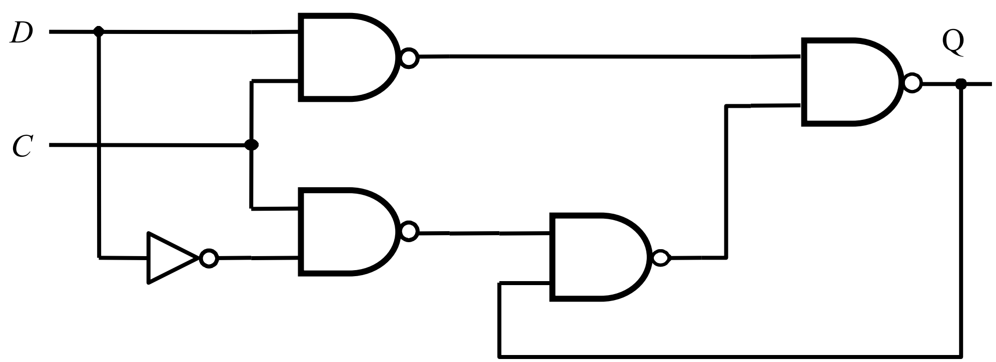
Mealy



Example circuits

SR latch and gated D latch





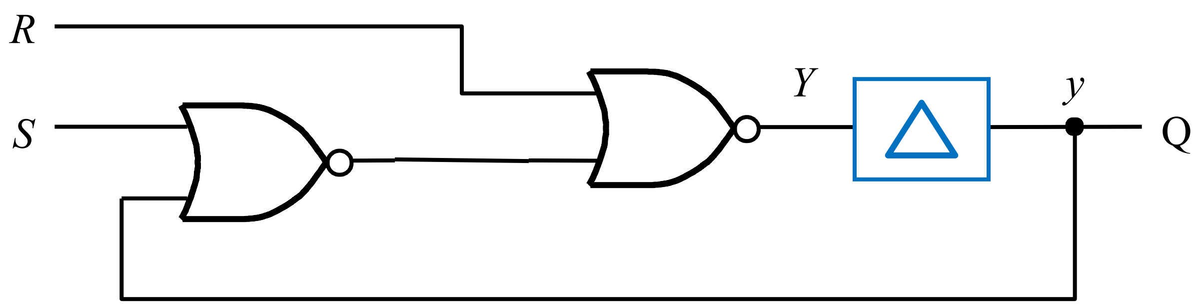
Important assumption

- A circuit is in *stable state* when all internal signals stop changing
- A circuit is operating in *fundamental mode* if the following restrictions hold
  - Only *one* input is allowed to change at a time
  - The input changes only after the circuit is *stable*
- We will assume the fundamental mode of operation for our circuits

Modeling gate delay

- Practical logic gates have propagation delays
- To simplify the analysis of an ASC, we will assume that the gates offer no delays. This will however be compensated by including a hypothetical delay element on the feedback path, allowing us to separate current and next states (see below)

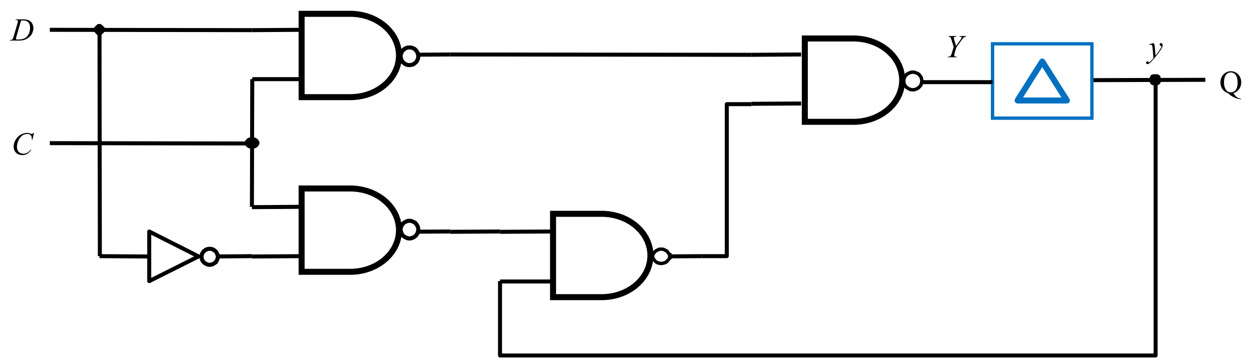
SR latch with modeled gate delay



Note that lowercase y corresponds to current state variable and uppercase Y next state variable

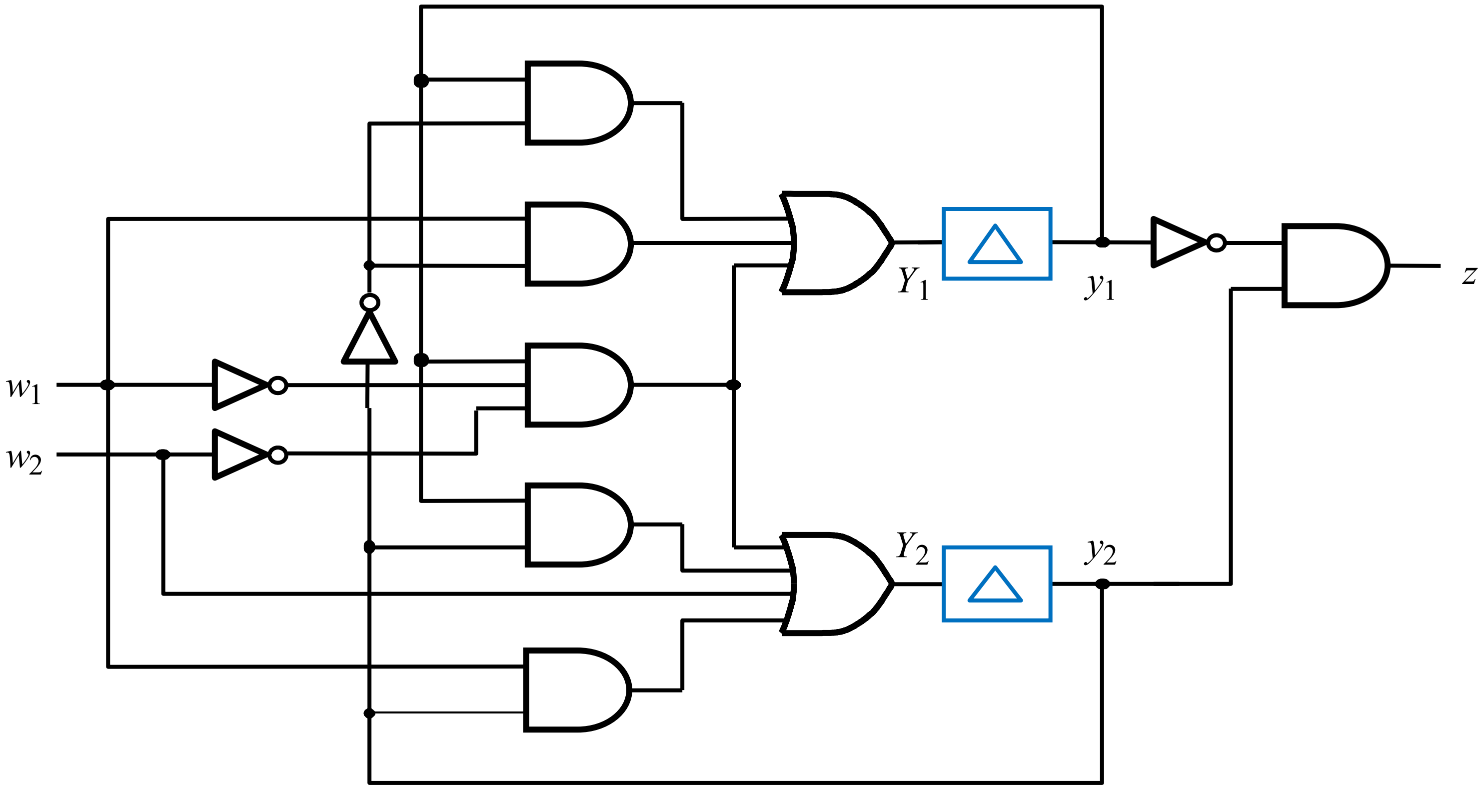
Modeling gate delay (contd.)

Gated D latch with modeled gate delay



Modeling gate delay (contd.)

A larger circuit:



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Lecture Slides: **Set 7 - Part 2**

Contents:

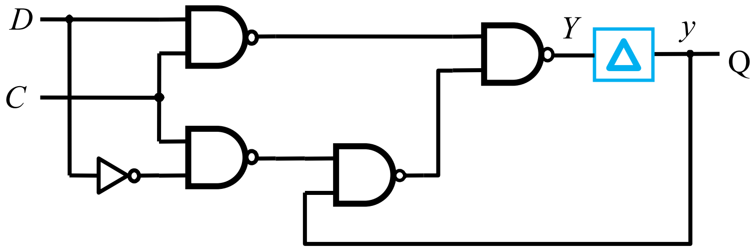
- Analysis of asynchronous sequential circuits

Analysis of asynchronous sequential circuits

- It is similar to that of synchronous seq. circuits (SSC)
- Given an ASC, basic steps for analysis are:
  1. Write logic expressions for the next-state (Y) and the output (z) in terms of the current state (y) and the input (w)
  2. Construct the excitation table (a.k.a. transition table)
  3. Obtain a flow table (similar to state table for SSC)
  4. Draw a state diagram if desired

Analysis of gated D latch

Gated D latch circuit is at right (along with a hypothetical delay element):



Looking at the circuit and using the consensus property, we can write:

$$Y = CD + C'y + Dy = CD + C'y$$

Analysis of gated D latch (contd.)

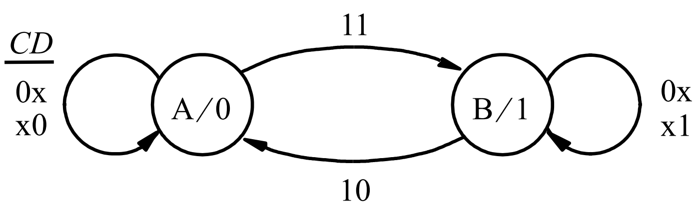
$$Y = CD + C'y + Dy = CD + C'y$$

Present state y	Next state CD = 00	01	10	11	Q
	Y	Y	Y	Y	
0	0	0	0	1	0
1	1	1	0	1	1

Excitation table (stable states are circled)

Present state	Next state CD = 00	01	10	11	Q
A	A	A	A	B	0
B	B	B	A	B	1

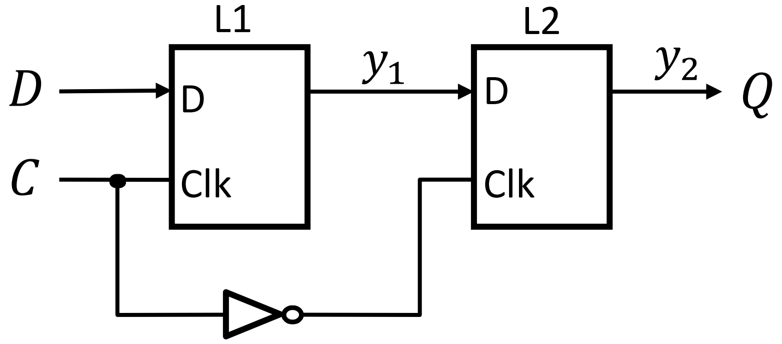
Flow table (obtained by replacing bit patterns of states with symbols)



State diagram

Analysis of DFF

Consider the DFF circuit is at right (hypothetical delay elements and next state variables (Y_1 and Y_2) are inside the latches (see two slides back)):



$$Y_1 = CD + C'y_1$$

$$Y_2 = C'y_1 + Cy_2$$

Analysis of DFF (contd.)

$$Y_1 = CD + C'y_1$$

$$Y_2 = C'y_1 + Cy_2$$

Present state $y_2\ y_1$	Next state $Y_2\ Y_1$, CD = 00	01	10	11	Output Q
00	00	00	00	01	0
01	11	11	00	01	0
10	00	00	10	11	1
11	11	11	10	11	1

Excitation table

Present state	Next state, CD = 00	01	10	11	Output Q
S1	S1	S1	S1	S2	0
S2	S4	S4	S1	S2	0
S3	S1	S1	S3	S4	1
S4	S4	S4	S3	S4	1

Flow table

Analysis of DFF (contd.)

Present state	Next state CD = 00	01	10	11	Output Q
S1	S1	S1	S1	S2	0
S2	S4	S4	S1	S2	0
S3	S1	S1	S3	S4	1
S4	S4	S4	S3	S4	1

Flow table (from previous slide)

Present state	Next state CD = 00	01	10	11	Output Q
S1	S1	S1	S1	S2	0
S2	–	S4	S1	S2	0
S3	S1	–	S3	S4	1
S4	S4	S4	S3	S4	1

Flow table with unspecified entries

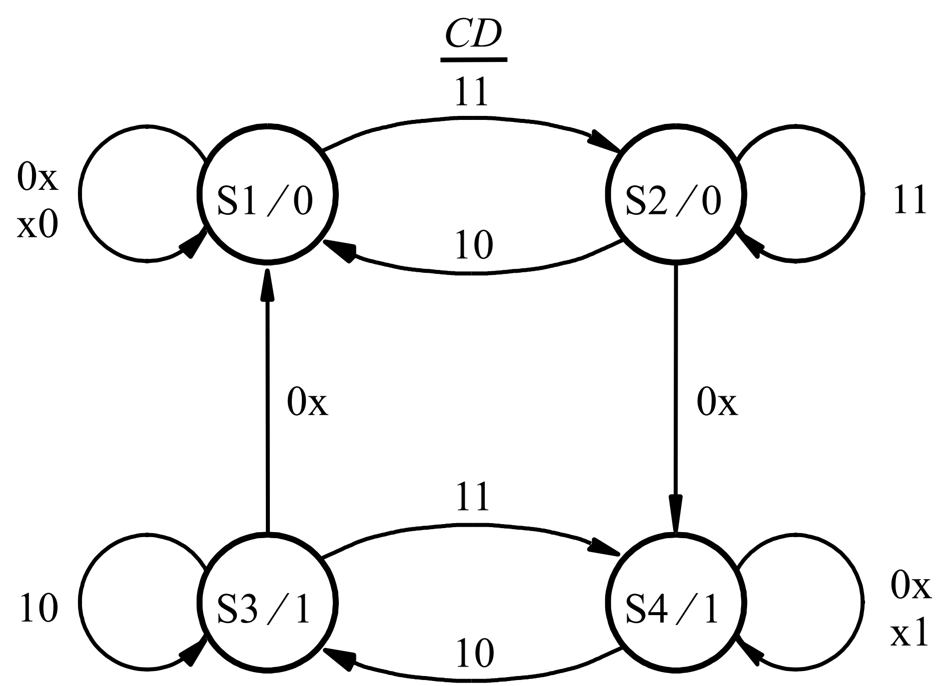
<u>Rationale</u>

- State S2 is stable with CD=11.
- In the fundamental mode, inputs must change one at a time. Hence CD will not change from 11 to 00 in one step and the corresponding transitional next state is removed/unspecified (see the table at right)
- Similarly, we can have an unspecified entry in row S3

Analysis of DFF (contd.)

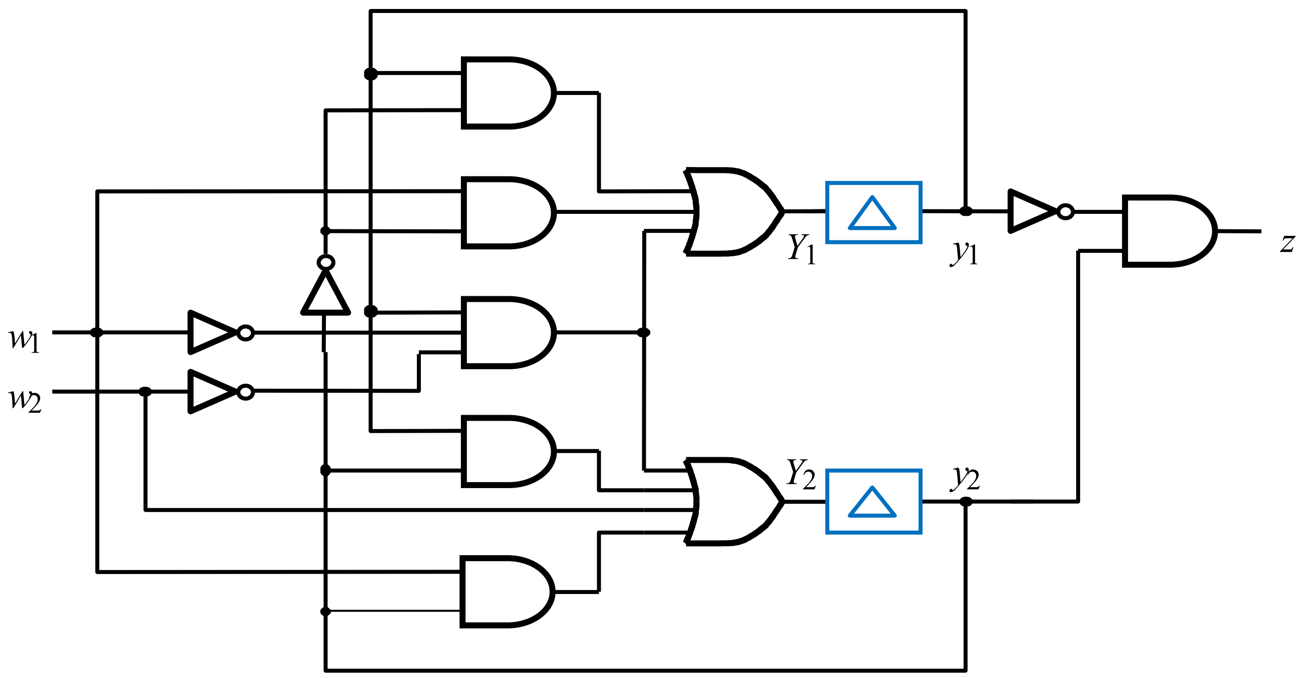
Present state	Next state				Output Q
	CD = 00	01	10	11	
S1	S1	S1	S1	S2	0
S2	–	S4	S1	S2	0
S3	S1	–	S3	S4	1
S4	S4	S4	S3	S4	1

Final flow table (from previous slide)



State diagram for D flip-flop

Analysis of a larger ASC



Logic expressions for the next state and the output are

$$Y_1 = y_1 y_2' + w_1 y_2' + w_1' w_2' y_1$$
$$Y_2 = y_1 y_2 + w_1 y_2 + w_2 + w_1' w_2' y_1$$
$$z = y_1' y_2$$

Analysis of a larger ASC (contd.)

Logic expressions from previous slide:

$$Y_1 = y_1 y_2' + w_1 y_2' + w_1' w_2' y_1$$
$$Y_2 = y_1 y_2 + w_1 y_2 + w_2 + w_1' w_2' y_1$$
$$z = y_1' y_2$$

Present state $y_2 y_1$	Next state $w_2 w_1 = 00$ $Y_2 Y_1$	01 $Y_2 Y_1$	10 $Y_2 Y_1$	11 $Y_2 Y_1$	Output z
00	00	01	10	11	0
01	11	01	11	11	0
10	00	10	10	10	1
11	11	10	10	10	0

Excitation table

⇒

Present state	Next state $w_2 w_1 = 00$	01	10	11	Output z
A	A	B	C	D	0
B	D	B	D	D	0
C	A	C	C	C	1
D	D	C	C	C	0

Flow table

Analysis of a larger ASC (contd.)

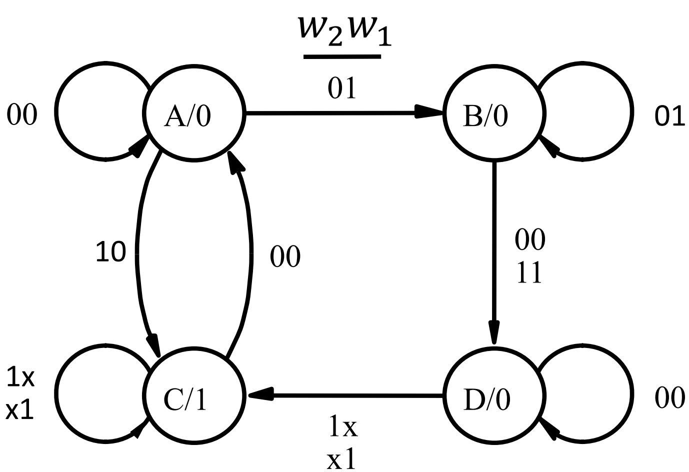
Present state	Next state w_2w_1 = 00	01	10	11	Output z
A	A	B	C	D	0
B	D	B	D	D	0
C	A	C	C	C	1
D	D	C	C	C	0

Flow table (from previous slide)

Present state	Next state w_2w_1 = 00	01	10	11	Output z
A	A	B	C	–	0
B	D	B	–	D	0
C	A	C	C	C	1
D	D	C	C	C*	0

Modified flow table

(* to allow $B \rightarrow D \rightarrow C$ *with* input $01 \rightarrow 11$)



State diagram

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Lecture Slides: **Set 7 - Part 3**

Contents:

- Synthesis of asynchronous sequential circuits

Synthesis of asynchronous sequential circuits

- It is the reverse task of analysis and has the same basic steps used for synchronous seq. circuits (SSC)
- Given a system specification, the basic steps for synthesis of an ASC are:
  1. Devise a state diagram
  2. Derive a flow table (and reduce the number of states if possible)
  3. Perform the state assignment and derive the excitation table
  4. Obtain the next-state and the output expressions
  5. Construct a circuit to implement these expressions

Serial parity generator

Specification (given):

- The system has an input w and an output z
- When pulses are applied to w
  - the output z is equal to 0 if the number of previously applied pulses is even
  - the output z is equal to 1 if the number of previously applied pulses is odd

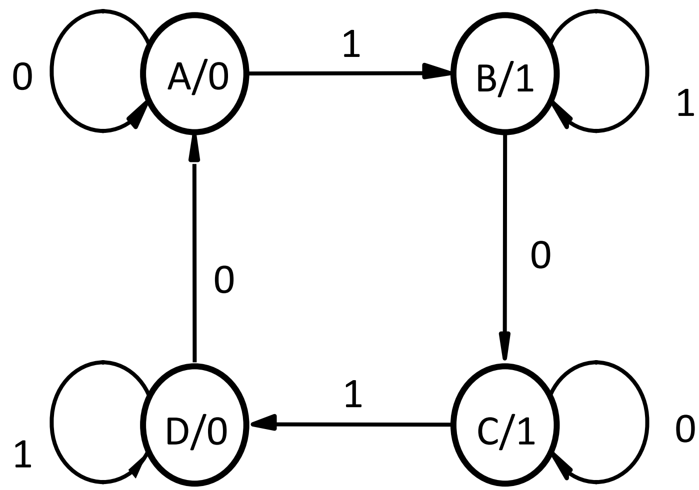
Assume a Moore machine

Serial parity generator (contd.)

Devising a state diagram:

- To start with, let A be the state that indicates that an even number of pulses has been received. I.e., state A produces z=0. As long as, w=0, the machine stays in A. Thus, A is stable with w=0
- When the next pulse arrives, i.e., w=1, the FSM moves to a new state, say B, which produces z=1. As long as, w=1, the FSM must remain stable be in B
- The next change in input is w=0 (i.e., completion of the current pulse), which causes the FSM move to another state, say C, that produces z=1. State C must be stable w=0
- When the next pulse arrives, i.e., w=1, the FSM moves to a new state, say D, which produces z=0. As long as w=1, the FSM must remain stable be in D
- The next change in input is w=0 (i.e., completion of the current pulse), the FSM moves to state A

Serial parity generator (contd.)



State diagram

Present State	Next state		Output
	$w = 0$	$w = 1$	z
A	A	B	0
B	C	B	1
C	C	D	1
D	A	D	0

Flow table

Serial parity generator (contd.)

State assignments:

- Let A=00, B=01, C=10, and D=11.

  This is a poor assignment since a transition from D=11 to A=00 (the last row of excitation table I) will cause both y_2 and y_1 to change, which is unlikely to happen at the same time. Depending on which variables changes first (racing), the destination will be different.
- A good assignment is: A=00, B=01, C=11 and D=10 (see excitation table 2)

Present state $y_2 y_1$	Next state $w = 0$		Output z
	$Y_2 Y_1$	$w = 1$	
00	00	01	0
01	10	01	1
10	10	11	1
11	00	11	0

Excitation table 1

Present state $y_2 y_1$	Next state $w = 0$		Output z
	$Y_2 Y_1$	$w = 1$	
00	00	01	0
01	11	01	1
11	11	10	1
10	00	10	0

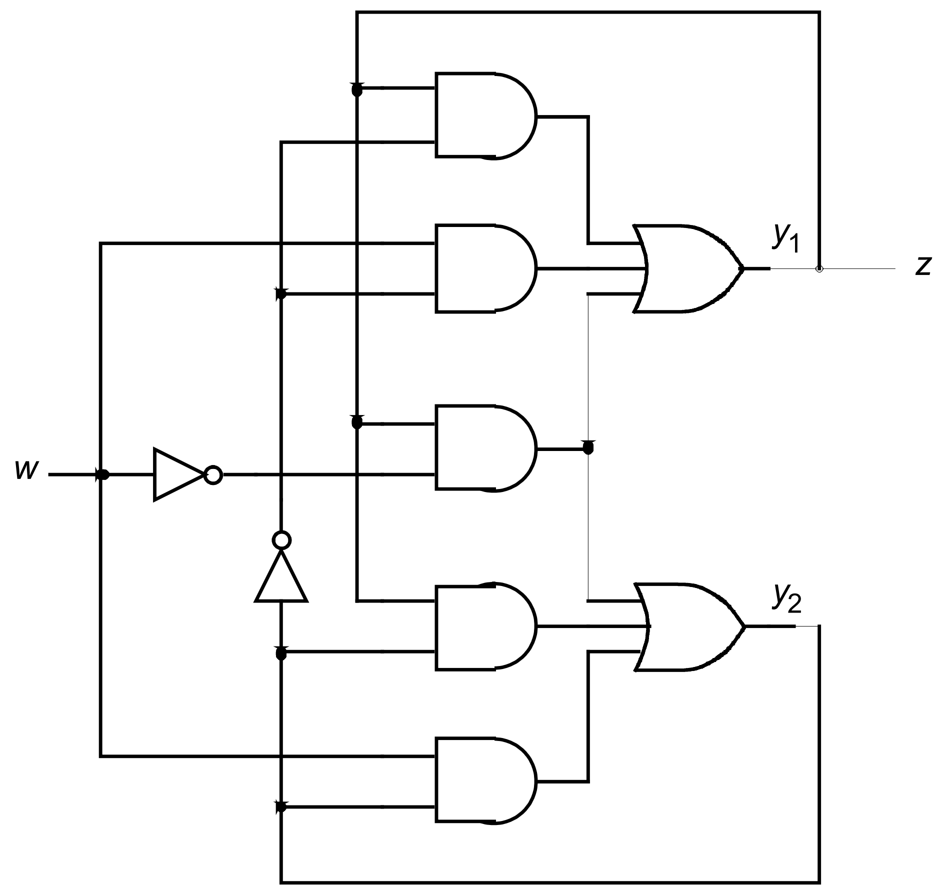
Excitation table 2

Serial parity generator (contd.)

From excitation table 2:

$$Y_1 = wy_2' + w'y_1 + y_1y_2'$$
$$Y_2 = wy_2 + w'y_1 + y_1y_2$$
$$z = y_1$$

The circuit is shown at right



Modulo-4 counter

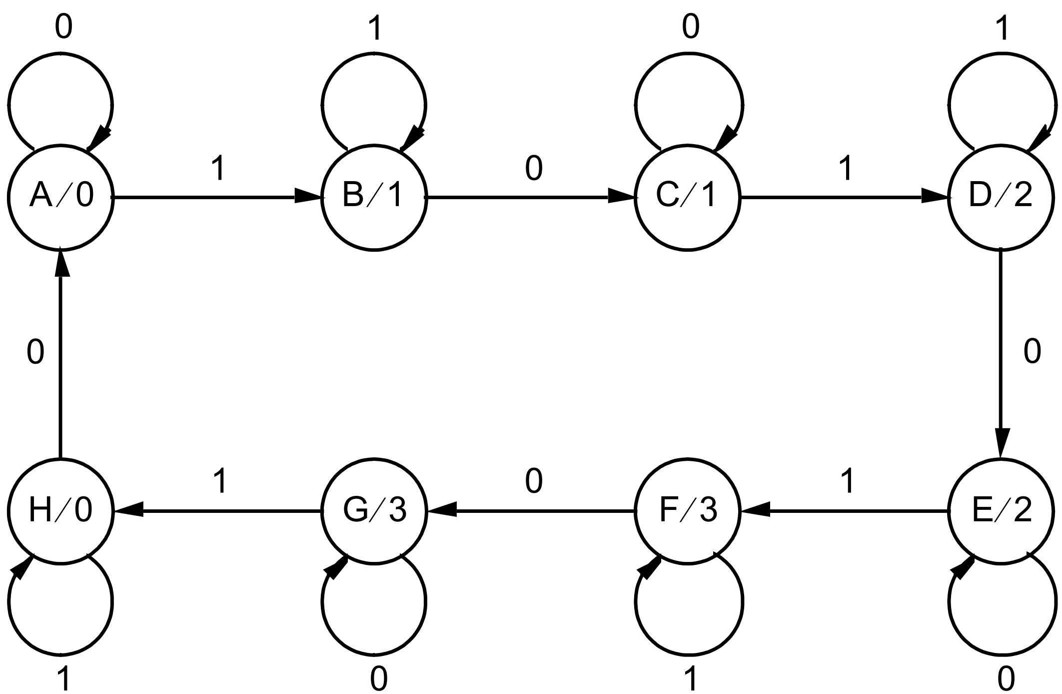
Specification (given):

- The system has an input w and an output $z = (z_2, z_1)$
- When pulses are applied to w
  - the output z is equal to i mod 4 if the number of previously applied pulses is i

Assume a Moore machine

Modulo-4 counter (contd.)

State diagram



Flow table

Present state	Next state		Output z
	$w = 0$	$w = 1$	
A	A	B	0
B	C	B	1
C	C	D	1
D	E	D	2
E	E	F	2
F	G	F	3
G	G	H	3
H	A	H	0

Modulo-4 counter (contd.)

Excitation table

Present state $y_3 y_2 y_1$	Next state $w = 0$ $Y_3 Y_2 Y_1$	Next state $w = 1$ $Y_3 Y_2 Y_1$	Output $z_2 z_1$
000	000	001	00
001	011	001	01
011	011	010	01
010	110	010	10
110	110	111	10
111	101	111	11
101	101	100	11
100	000	100	00

Logic equations for Y's and z's
(Yours to try)